Chapter 3 - Partial Differential Equations - Notes

Jordan Tucker

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1 A Mathematical Representation of Waves

1.1 Introduction and examples

If a function u is the value of some measurement of a one dimensional medium that depends on position x and time t,the partial derivatives of u with respect to x and t have important physical meanings.

A **partial differential equation** (PDE) for a function $u(x,t)$ is a differential equation that involves one or more of the partial derivatives of u with respect to x and t. We will usually denote the partial derivatives of u by

$$u_t = \frac{\partial u}{\partial t}, \quad u_x = \frac{\partial u}{\partial x}, \quad u_{xt} = \frac{\partial^2 u}{\partial t \partial x}, \quad \dots$$

Common PDEs with solutions illustrating waves:

(a) Advection equation: $u_t + cu_x = 0$.

(b) Diffusion equation: $u_t = Du_{xx}$.

(c) The linearized Burgers equation: $u_t + cu_x = Du_{xx}$.

(d) Burgers equation: $u_t + uu_x = Du_{xx}$.

(e) Inviscid Burgers equation: $u_t + uu_x = 0$.

(f) Wave equation: $u_{tt} = c^2 u_{xx}$.

(g) Korteweg-deVries equation: $u_t + uu_x + u_{xxx} = 0$.

1.2 An intuitive view

At a fixed position x in the medium, the value of $u_t(x,t)$ gives the rate at which u is changing with respect to time. A positive value of $u_t(x,t)$ indicates that at position x, u is increasing as time increases. A negative value of $u_t(x,t)$ indicates that u is decreasing at position x as time increases. In applications where $u(x,t)$ denotes a displacement at position x, $u_t(x,t)$ and $u_{tt}(x,t)$ have the particular interpretation of velocity and acceleration.

At a fixed time t, the values of $u_x(x,t)$ and $u_{xx}(x,t)$ provide information about the slope and concavity of the graph of $u(x,t)$ as a function of x. In particular applications these represent quantities such as flux and stress.

Example: Suppose u represents the temperature on a rod at position x and time t. Suppose the temperature of the rod is governed by the heat equation:

$$u_t = Du_{xx} \quad (D > 0 \quad constant),$$

u_t is directly proportional to the concavity of $u(x,t)$. If $u(x,t)$ is concave up, u_t is positive so the temperature will increase. If $u(x,t)$ is concave down, u_t is negative so the temperature will decrease. This makes sense because heat flows from hotter regions to colder regions.

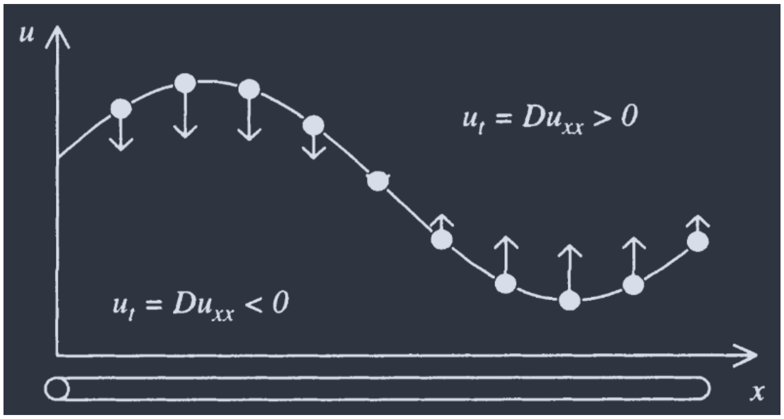

Figure 1: According to the heat equation, the change in temperature u at each point x depends on the concavity of the temperature profile at that instant.

1.3 Terminology

PDEs can be classified many different ways.

The **order** of a PDE is the order of the highest order partial derivative in PDE.

A PDE is **first order linear** if it follows the form

$$a_1 u_t + a_2 u_x + bu = f \tag{1}$$

Where a_1 and a_2 are not both 0, and a_1, a_2, b, and f are constants or functions of x and t (but cannot depend on u). If a first order PDE cannot be written in this form, then it is **nonlinear**.

A PDE is **second order linear** if it can be written in the form

$$a_{11} u_{tt} + a_{12} u_{tx} + a_{22} u_{xx} + b_1 u_t + b_2 u_x + cu = f \tag{2}$$

Where a_{11}, a_{12}, and a_{22} are not all zero, and a_{ij}, b, c, and f are constants or functions of x and t (but don't depend on u). If a second order PDE does not follow this form, then it is **non-linear**.

We can typically classify linear equations and **homogeneous**. If $f = 0$, then the PDE is **homogeneous**. If $f \neq 0$, then the PDE is **nonhomogeneous**. We typically do not classify nonlinear PDEs as homogeneous or nonhomogeneous.